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MATHEMATICS
Paper 1
July – August 2026
2 ¼ Hours



UGANDA MUSLIM TEACHERS' ASSOCIATION
UMTA JOINT MOCK EXAMINATIONS – 2026

Uganda Certificate of Education

MATHEMATICS

Paper 1

2 Hours 15 Minutes

INSTRUCTIONS TO CANDIDATES:

- *This paper consists of two sections; A and B. It has six examination items.*
- *Section A has two compulsory items.*
- *Section B has two parts; I and II. Respond to one item from each part.*
- *Respond to four examination items in all.*
- *Any additional item(s) answered will not be scored.*
- *All responses must be written in the answer sheet(s) provided.*
- *A Graph paper should be provided.*
- *Silent non-programmable scientific calculators and mathematical tables with a list of formula may be used.*

SECTION A

Respond to both items in this section.

Item 1.

Your school has contracted a company to install a solar power system in order to address the issue of frequent hydro-electricity blackouts. The project will cost the school a total of one billion two million forty-eight thousand Uganda shillings. The cost is to be paid in two instalments in a ratio of 7:10 respectively. The first instalment will be deposited on the company's bank account. Three quarters of the second instalment amount will be paid in cash on completion of the installation of the system.

The Environmental Protection Board of the Local Council will approve the project if the crystalline solar cell peak efficiency factor exceeds a threshold of $15\sqrt{2}$. Given that crystalline solar cell to be installed has a peak efficiency factor

expressed as, $\frac{\sqrt{72} \times 2^5 \times 4^{-1}}{8^3 \times 16^{-2}}$ they need to know whether the project will be

approved. Of the total land allocated for installation, the Solar Arrays will take up

0.35, the Inverter Stations $\frac{3}{20}$, and the Battery Storage & Substation $\frac{2}{5}$. They need

to know whether the remaining percentage will be enough for the Green Buffer Zone, given that it requires at most 12% of the total land.

Two types of inverters will be installed, and they will require to be serviced.

Type-A inverters will be serviced every 18 days, while **Type-B** inverters will be serviced every 30 days. If both are to be serviced on the day of installation, they need to know in how many days both inverters will be serviced again.

Task:

(a) How much will the company receive in cash on completion of the contract.

(b) Will the project be approved by the Environment Protection Board?

Justify

(c) Will the remaining percentage of land be enough for the Green Buffer Zone?

(d) In how many days will both inverters be serviced?

Item 2.

Your family plans to purchase a total of 30 Electric and Petrol motorcycles for delivering products to their customers. The total daily maintenance budget to be allocated for these motorcycles is \$420. Given that the daily maintenance cost for an Electric motorcycle is \$20 and that of a Petrol motorcycle costs \$8, they need to determine the number of motorcycles of each type that they are to buy. To create parking space for the motorcycles, your family plans to buy a rectangular piece of land. The length of the land must be 4 meters longer than twice its width. The total area required for parking of motorcycles is 160 square meters.

Task:

(a) How many motorcycles of each type will they buy?

(b) What are the dimensions of the space they will buy for parking the motorcycles?

SECTION B

This Section has two Parts: I and II.

Part I

Respond to one item from this part.

Item 3.

Districts A, B and C were affected by an earthquake. The rescue team organized a given number of crates of food, Medical and water purification supplies to be delivered to the affected districts. Given that 15 crates contain only medical supplies, 40 crates contain Food and some other supplies, 38 crates contain Water Kits and some other supplies, 30 crates contain medical supplies and some other supplies, 12 crates contain both Food and Medical Supplies with some water purifiers, 14 crates contain both Food and Water Kits with some medical supplies, 11 crates contain both Medical Supplies and Water purification Kits with some food supplies and some crates contain all the three Supplies, they need to sort all the mixed crates before they start distributing them. To improve the welfare of the victims, the rescue team plans to include additional hygiene kits if the number of crates containing at most two types of supplies is greater than 60%.

Those to distribute the items are to make 2 trips to District A, 3 trips to District B and 2 trips to District C. The rescue team needs to know how much to prepare for distribution given that each trip to A will cost Shs.88,000 in terms of fuel, each trip to B will cost Shs.100,000 and each trip to C will cost Shs.70,000.

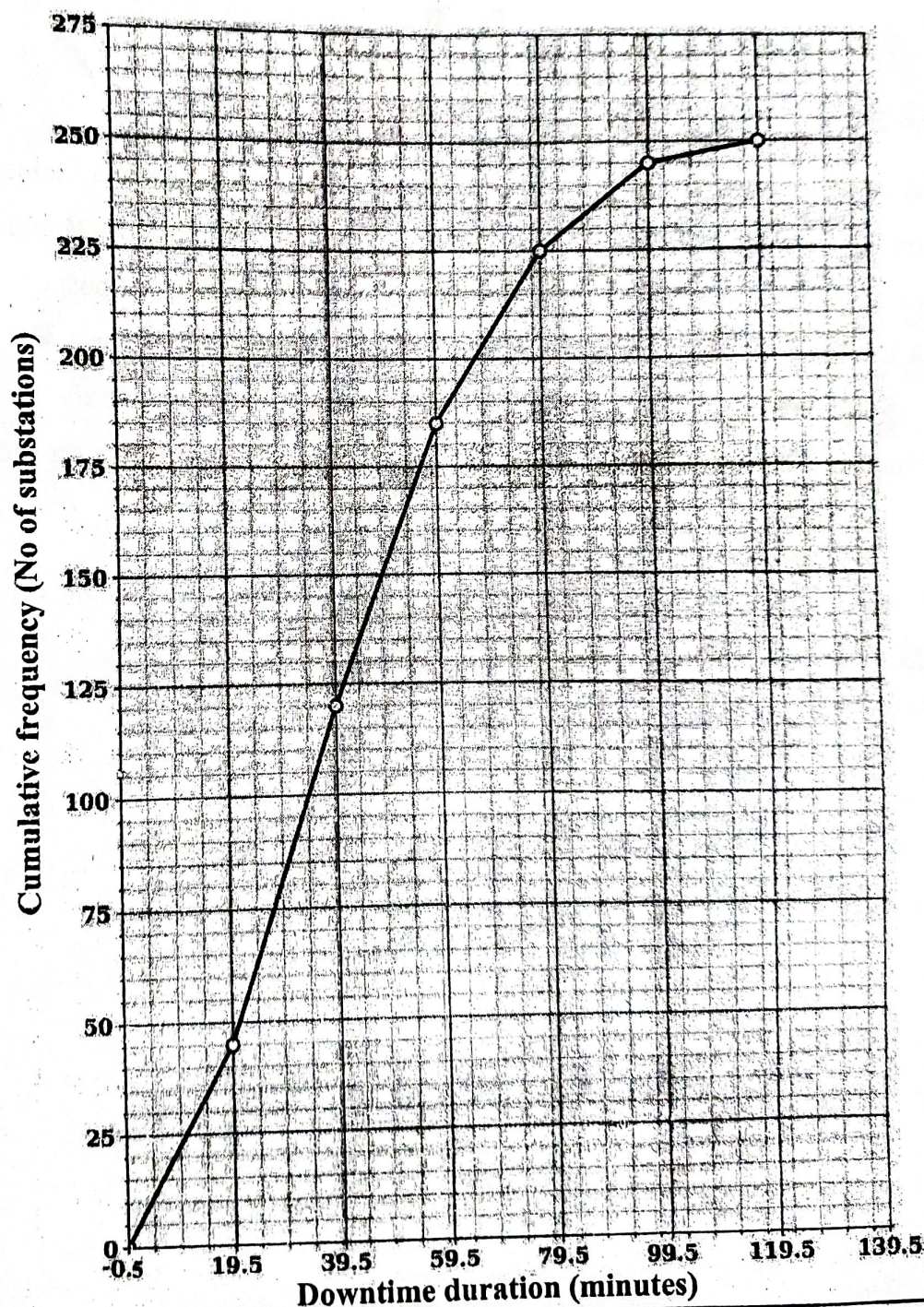
Task:

- (a) With aid of a relevant presentation, determine whether the rescue team will include additional hygiene kits.
- (b) By arranging data in rows and columns while stating the orders of arrangements, conclude about the amount they will need for all the trips to be made.

Item 4.

The Capital City Council proposed to switch from diesel to Electric buses. The higher officials are to approve the proposal if within 90 percentage of the substations, the duration of grid voltage fluctuations exceed 75 minutes.

On requesting for data about the duration of grid voltage fluctuations from different substations. The council was provided a graph below:



There was also need to report about the duration of grid voltage fluctuations for majority of the substations that were sampled.

Task:

- (a) Conclude about the duration of grid voltage for majority of the substations that were sampled.
- (b) Will the higher officials approve the proposal? (Justify your answer)

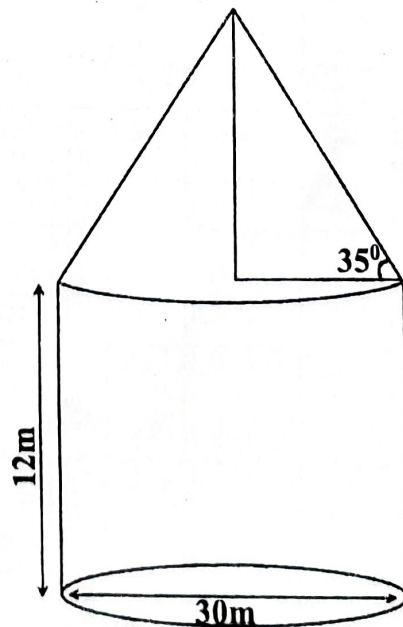
Part II

Respond to one item from this part.

Item 5.

Your town Council has acquired a large piece of land to develop a community resort. The survey team has mapped three primary control points: **A**, **B** and **C**, using a local grid system and compass bearings with reference to the post office, **P**. **A** is **6 km East** and **8 km North** of **P**. **B** is **14 km east** and **2 km North** of **P**. **C** is **5 km West** and **7 km North** of **P**. The council wants to know the direct distances and bearings of **B** and **C** from **A** to include them in the resort's directory book.

The central pavilion of the resort features a hollow right circular cone sitting perfectly on top of a solid cylinder base as illustrated below:



The exterior walls of the pavilion must be covered with weatherproofing, and the inside of the entire pavilion must be fitted with an advanced centralized climate control system. The council wants to know the area that needs weather proofing and the inside volume of the pavilion.

Task:

- (a) What should the council include in the resort's directory about A, B and C.
- (b) Calculate the,
 - (i) area that needs weatherproofing.
 - (ii) volume of air space inside the combined cylinder-and-cone pavilion structure.

Item 6.

There is an existing resort whose board of directors wants to expand by adding a new camping zone on a quadrilateral piece of land **ABCD**. From point **A**, the boundary runs **60 m** on a bearing of **045°** to Point **B**, then **80 m** on a bearing of **135°** to Point **C**. The board wants to place a security outpost exactly equidistant from the two boundary lines **AB** and **BC**, and exactly equidistant from points **A** and **C**. Part of the camping zone will feature a triangular activity field, **ACD** such that **CD = 110 m** and angle **ACD 80°**. There is need to run an underground power cable directly from **A** to **D** and each meter of the cable costs **Shs.166,500**. The board wants to know the direct distance and bearing of the security outpost from point **A**, and the cost of the power cable.

The expansion project will be managed by a qualified quantity surveyor on a monthly gross pay of **Shs.11,523,000** subject to pay as you earn (PAYE) computed as follows.

Amount (Shs.)	0 – 335,000	335,001 – 410,000	410,001 – 485,000	485,001 – 10,000,000	Above 10,000,000
PAYE (%)	0	10	20	30	40

The board wants to know the surveyors' net pay.

Task:

- (a) Using construction,
 - (i) find the actual distance and bearing of the security outpost from point A.
 - (ii) calculate the exact cost of the power cable AD.
- (b) Calculate the Surveyor's net monthly pay.

END